Transport-coupled evaluation of river-network CO_2 closures: Conditional process-allocation diagnostics from East River concentration observations

Author-review version, 27 September 2026. This revision reconciles the manuscript with the archived numerical results and distinguishes verified calculations from unresolved methodological issues. It is not a submission-ready validation of the spatial support or sparse-closure evaluation. The original observations and production outputs have not been altered.

Authors: To be completed (pending)

Affiliations: To be completed (pending)

Key Points

• Public East River observations (Saccardi and Winnick) were reanalysed using a logical routing tree and a separate NHDPlus HR mapping and filtering network; the main residual learners use fold-local targets.

• Under grouped leave-one-reach-out transport CV, the Residual-AI MLP RMSE is 0.0372 mol m^{-3} versus Baseline 0.0486, but the random forest is worse (0.1011) and flux diagnostics remain unstable sample-support quantities.

• A local mass balance admits source–exchange compensation; a conditional profile SSE describes this trade-off under fixed upstream concentrations, not general network identifiability.

Plain Language Summary

River-carbon models represent downstream transport, carbon inputs and exchange with the atmosphere. A concentration measurement constrains their combined effect, but does not necessarily separate those effects. We reanalysed 120 selected samples from the publicly archived East River campaign of Saccardi and Winnick (2021). The measurements were collected by that study, not by the present authors. Our calculations compare a baseline transport model with rules that supply a learned source term or adjust the gas-exchange rate. A simplified routing tree carries the forward calculation; a separate public hydrographic network supplies mapping and spatial-filtering information. For the main learners, training targets were rebuilt without the concentrations in the held-out reach. The multilayer perceptron gave a lower sample-weighted concentration error than the baseline, whereas the random forest did not. The gain was uneven across reaches, and the calculated flux diagnostics differed greatly. Those diagnostics overlap in spatial support and are not watershed totals or independently measured fluxes. A separate error-profile calculation shows compensation between a source term and gas exchange when upstream concentrations are fixed. This is a conditional explanation of the model balance, not proof that all concentration datasets fail to distinguish processes. Unresolved spatial support and numerical clipping still limit the empirical comparison. The case therefore motivates checking how a model obtains a good concentration fit, alongside checking the fit itself.

Abstract

River-carbon models combine transport, unresolved sources and sinks, and air–water gas exchange. Concentration error measures their combined prediction, not the validity of each process allocation. We reanalysed 120 publicly archived East River observations (Saccardi and Winnick, 2021), grouped into eight logical reaches. Forward transport uses a simplified routing tree, USGS outlet discharge and published synoptic tributary discharge; a separate NHDPlus HR network supports mapping and filtering. We compare a zero-residual Baseline, learned residual source–sink closures, and a multiplicative gas-transfer correction. The main learners rebuild residual or transfer-velocity targets within each reach-held-out fold. Sample-weighted concentration RMSE was 0.0486 mol m^{-3} for the Baseline, 0.0372 for the multilayer perceptron (MLP), and 0.1011 for the random forest. The MLP gain was concentrated in one three-sample reach, and three held-out MLP concentrations were clipped to zero. Overlapping sample-area flux diagnostics were 1.21×10^{3}, 1.94×10^{6} and 7.28×10^{5} mol d^{-1}, respectively; they are not watershed totals. The gas-transfer correction did not lower RMSE, and 60% of full-data inverse targets lacked a positive transfer-only solution. A conditional deterministic profile SSE, holding diagnosed upstream concentrations fixed and sharing one transfer scale and source across samples, gave RMSE 0.0360–0.0396 mol m^{-3} over transfer scales 0.01–1. Its single-sample zero valley is an algebraic property, not a network identifiability proof. Resolved filter-scale results remain sampling-conditional, while spatial-support inconsistencies and the separate sparse branch require correction. The archived experiment illustrates why concentration skill and process-allocation diagnostics should be reported separately; it does not establish a validated transferable closure.

Keywords: river carbon cycling; environmental model evaluation; subgrid closure; gas-transfer velocity; grouped cross-validation; practical equifinality

1. Introduction

Rivers link terrestrial, aquatic, and atmospheric carbon cycling by transporting dissolved carbon, transforming it through biological and biogeochemical processes, and exchanging CO_2 with the atmosphere. Riverine CO_2 evasion is a substantial component of inland-water carbon budgets, but both the magnitude and dominant controls of that flux vary across river networks and over time (Hotchkiss et al., 2015; Gómez-Gener et al., 2021; Battin et al., 2023). Predictive models must therefore represent several processes that operate simultaneously: downstream advection, lateral and internal carbon inputs, in-stream transformation, and air-water gas exchange. At the reach scale, however, observations and model states average over spatial heterogeneity that may occur over much shorter distances. A reach-scale model can consequently reproduce concentration while compensating for unresolved source or sink processes through another term in the mass balance. This possibility matters whenever model performance is judged mainly from concentration observations.

The East River watershed provides a useful setting in which to examine this problem because a process-based stream-network CO_2 model has already been developed for the system. Saccardi and Winnick (2021) represented downstream transport together with CO_2 sources and atmospheric exchange to predict spatial variation in stream CO_2 concentrations and fluxes. Their work illustrates how several process terms jointly determine the same downstream concentration state. Gas exchange is commonly represented through an empirical transfer velocity, and Raymond et al. (2012) related gas-transfer velocity to hydraulic properties of streams and small rivers. Such parameterizations make gas exchange operational at network scale, but they do not remove uncertainty in the other terms of the carbon balance. Groundwater inputs, lateral inflows, metabolic heterogeneity, and other sub-reach processes may remain incompletely observed or resolved. Errors in these processes and errors in gas exchange can therefore enter the same concentration equation and potentially offset one another.

This compensation complicates model evaluation because different process allocations can produce similar concentration responses. Bennett et al. (2013) argued that environmental-model performance should be characterized using evidence appropriate to model purpose rather than a single goodness-of-fit statistic. Vilas et al. (2023) likewise treated model-data discrepancy as information that may arise from the model, the observations, or their interaction, while Markovich et al. (2022) evaluated

alternative methods explicitly in the presence of model error. For river-carbon transport, these considerations imply that a lower concentration error is not necessarily sufficient to identify which process representation is better supported. If an unresolved source-sink correction and the gas-transfer velocity can both alter the same predicted concentration, optimizing either term may reduce residual error. The relevant test is whether their effects remain distinguishable when each candidate closure is coupled back to the transport model and evaluated under reach-grouped transport coupling, subject to the boundary-conditioning and target-construction limitations described below.

Machine learning provides one possible representation of unresolved process terms. In climate modeling, for example, spatial coarse-graining has been used to define unresolved tendencies and to train data-driven subgrid parameterizations, including models tested across resolutions (Yuval & O'Gorman, 2020). Related work on data-driven closure of river-carbon transport is also in preparation (Gao et al., manuscript in preparation). The central difficulty is not whether a flexible model can fit a diagnosed residual in the data used for training, but whether that residual representation remains useful after transfer to a new spatial group and reinsertion into the governing balance. A residual may encode repeatable unresolved dynamics, but it may also absorb errors in hydraulic representation, boundary conditions, observations, or other model components. Strong in-sample fit therefore provides limited evidence for a closure unless its effect is evaluated through the coupled model under held-out conditions.

Here we examine this question with 120 public East River observations assigned to eight logical reaches. Three parts of the analysis serve different purposes. First, predictions from residual and gas-transfer closures are reinserted into the quasi-steady balance before held-out concentration is scored. Second, spatial filtering on an NHDPlus HR representation describes how a diagnosed source–sink residual, S_sgs, changes with aggregation. Third, an algebraic objective surface separates local source–exchange compensation from the predictive comparison. A sparse representation provides an additional descriptive experiment, but its archived evaluation does not yet meet the main protocol's fold-local target requirement. Our contribution is this explicit separation of prediction, residual diagnosis and conditional process allocation in a public-data case study. We do not claim that the balance equation, empirical transfer relation, public observations or algebraic non-uniqueness were newly discovered here. Unequal sampling, empirical geometry, unresolved spatial support and the absence of independent evasion measurements bound the conclusions.

2. Methods

2.1 Study data and river-network representation

The study reuses public observations from the upper East River watershed near Almont, Colorado (hydrologic unit code 14020001). The analysis selects 120 dated, model-included samples from the archived campaign table of Saccardi and Winnick (2021), collected between 2 and 11 August 2019. This number describes our selected analysis set, not necessarily the entire original campaign. Logical reaches R001–R008 contain 1, 3, 15, 24, 17, 1, 1 and 58 samples, respectively (Table 1). The three single-sample reaches enter the routing bookkeeping but provide limited evidence for transfer. These eight labels define evaluation groups; they are not eight statistically independent replicates of a catchment-scale experiment. The production routing tree contains seven tributary nodes and six mainstem segments, R008_S1–S6. Mainstem order and tributary attachment follow confluence-northing thresholds. This is a simplified logical tree, not a verified path-compressed representation of the fine NHD directed graph.

Three public sources supply distinct inputs. The HydroShare supplement provides campaign chemistry and synoptic tributary discharge. NHDPlus HR flowlines for HUC 14020001 provide a separate mapping and filtering network, whose directed topology uses endpoint matching and the flow-direction attribute. That fine network must not be substituted verbally for the logical tree used in forward transport. Mainstem outlet discharge comes from USGS gage 09112500 on the sample dates; tributary discharges use the published synoptic values without gage-ratio scaling. Sample identifiers join chemistry to logical reaches, projected locations assign samples to mainstem segments, and fine-flowline mappings support

maps and filter-cell membership. A geographical match establishes location, not automatically a connected, correctly oriented physical transport path. The archived implementation still uses flowline-local projected chainage and unbounded sample-midpoint support for its production control volumes. A common cumulative path coordinate and a fixed bounded domain therefore remain prerequisites for interpreting the comparison as a geographically verified transport experiment.

Channel width and depth are not measured along the corridor, so both are estimated from the hydraulic-geometry relations of Raymond et al. (2012), fitted to 1026 paired measurements in streams and small rivers: width W = 12.88·Q^0.42 (m), depth h = 0.4·Q^0.29 (m), and bulk velocity u = 0.19·Q^0.29 (m s^{-1}), with discharge Q in m^3 s^{-1}. These are model-derived hydraulic inputs rather than measured cross-section properties, and the width enters velocity, k_600, and the water-surface area A_s = L·W. All hydraulic and gas-exchange quantities are conditional on this adopted geometry; no independent cross-section survey is available for this corridor. Biogeochemical covariates are likewise incomplete: dissolved inorganic carbon (DIC) and dissolved organic carbon (DOC) are available for 41 of the 120 samples, and alkalinity, nitrogen, phosphorus, and photosynthetically active radiation were not available for this campaign. A same-day merge against the Water Quality Portal returned no matching samples (0 of 120), and the StreamPULSE database contains no East River sites. These gaps constrain the covariate set available to the closures. Figure 2 shows the logical-reach assignment and the distribution of the 120 campaign samples on the river network.

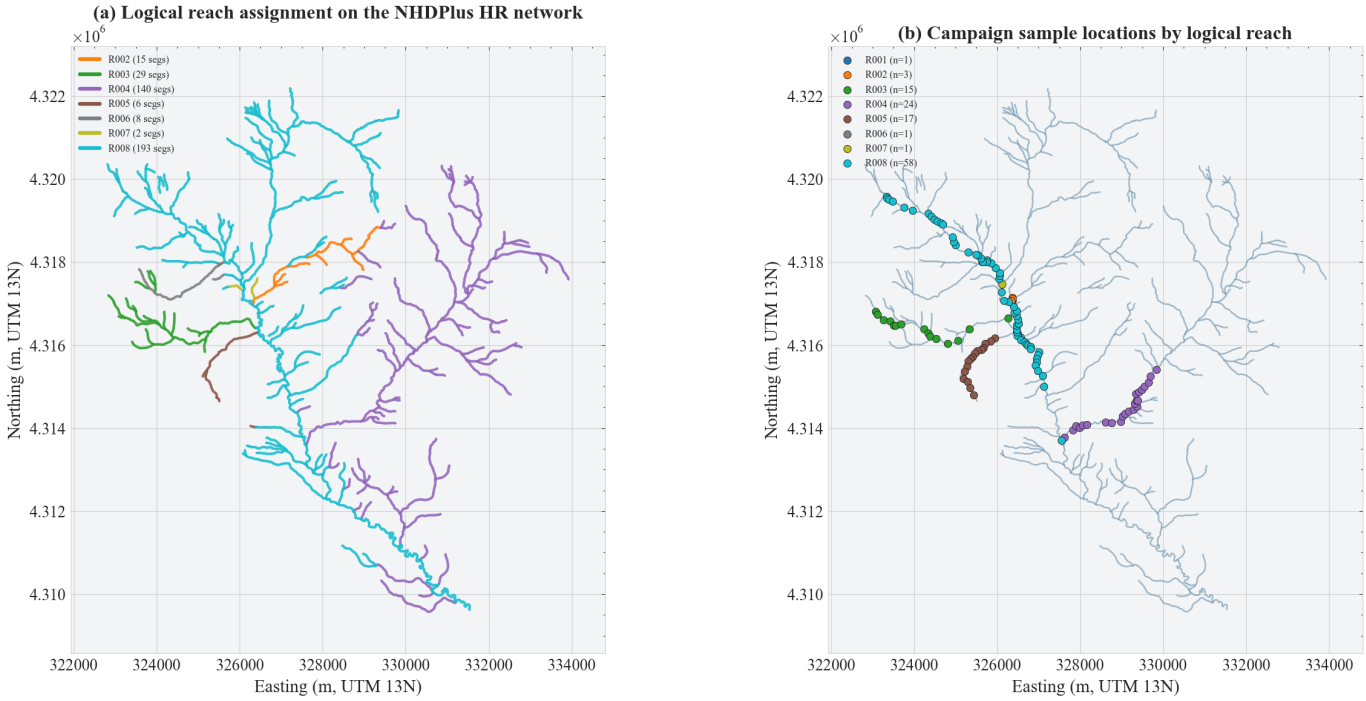


Figure 2. (a) Logical reach assignment on the NHDPlus HR network: segments were assigned to the logical study-reach labels by GNIS name matching and proximity to campaign coordinates; colors indicate reach assignment, and the single-sample reach R001 receives no assigned segment in this rendering. (b) Locations of the 120 campaign samples; point colors denote logical reaches R001–R008 and R008 contains 58 samples. The background provides spatial context only and is not used in the assignment.

2.2 Quasi-steady CO_2 mass balance and gas exchange

Each sample is associated with a model support length L and computed width W. In the archived production calculation, L comes from within-segment sample-midpoint spacing, with a 100 m fallback for a single sample, rather than a bounded tessellation of a verified reach domain. The resulting support is a current limitation, not a measured reach length. Carbon balance is evaluated quasi-steadily for each sampling date; this is not a time-integrated daily simulation. With equal inflow and outflow discharge within a control volume and no separate lateral term, advection, unresolved sources and atmospheric exchange satisfy

Eq. (1): Q(C_in - C) + (A_s/tau_d)[S_sgs - k(C - C_eq)] = 0

where Q is discharge ($m^3\ s^{-1}$), C_in and C are upstream and control-volume concentrations (mol m^{-3}), k is transfer velocity (m d^{-1}), and S_sgs is the areal source–sink term (mol $m^{-2}\ d^{-1}$). C_eq is atmospheric-equilibrium concentration, A_s = L·W is water-surface planform area (m^2), and tau_d = 86400 s d^{-1} converts daily areal flux to mol s^{-1}. Planform area is distinct from hydraulic cross-section area. The archived hydraulic inputs use the three empirical relations in Section 2.1: their product uWh = 0.97888Q is close to, but not exactly, Q. We therefore do not claim that the stored empirical u is exactly Q/(Wh); changing to a continuity-derived velocity would require updating k and the derived predictors. For residual diagnosis, C is observed and S_sgs is inferred. For forward simulation, the closure supplies S_sgs and C is solved. Dividing Eq. (1) by A_s/tau_d gives the daily areal form

Eq. (2): q_A(C_in - C) + S_sgs - k(C - C_eq) = 0

in which q_A = tau_d·Q/A_s (m d^{-1}) is a daily area-normalized discharge and every term has units mol $m^{-2}\ d^{-1}$.

All closures are inserted into this same balance: a closure configuration is defined entirely by how it supplies S_sgs and k, and every configuration is scored after the identical transport calculation is re-solved. Because the transport numerics are held fixed, differences among configurations arise from their closure specification and the resulting allocation of model discrepancy.

Gas exchange is summarized by the model flux density

Eq. (3): F_CO2 = k(C - C_eq)

The flux diagnostic sums F_CO_2 times the assigned water-surface area A_s, giving units mol d^{-1}. Samples can revisit or overlap spatial support, and the sum combines observations from different dates. It is therefore neither a watershed flux at one time nor a time-integrated campaign emission. It compares model allocations under the archived sample support; it is not an independently observed evasion quantity.

The empirical transfer velocity follows the energy-dissipation parameterization of Ulseth et al. (2019), which was the formulation used by the original East River model (Saccardi & Winnick, 2021) and which accounts for the high turbulence of steep mountain streams. The Schmidt-600-normalized velocity k_600 is estimated from the stream energy-dissipation rate e_D = g·S·u, and the CO_2-specific velocity is obtained by Schmidt-number scaling:

ln k600 = 1.18 ln eD + 6.43 (eD >= 0.02 m2 s-3); eD = g S u; k_emp = k600 (Sc/600)^-0.5

We distinguish the Schmidt-600-normalized velocity k_600 from the CO_2-specific empirical velocity k_emp. The high-energy branch (e_D ≥ 0.02 $m^2\ s^{-3}$) applies throughout the study corridor; the Schmidt number Sc is evaluated at the sample water temperature. Because the choice of k_600 parameterization affects every gas-exchange result, Raymond et al. (2012) Model-4 k_600 values are retained as a sensitivity column in the processed observation table (median k_600 ≈ 98 m d^{-1} versus ≈ 268 m d^{-1} for Ulseth); the primary evaluation below uses Ulseth throughout. The equilibrium concentration C_eq is computed from Henry's law with a temperature-dependent solubility constant. The constant follows the NIST/Sander compilation for CO_2 in water, k^°_H(298.15 K) = 0.035 mol kg^{-1} bar^{-1} with a van 't Hoff temperature dependence d(ln k_H)/d(1/T) = 2400 K, so C_eq = k_H(T)·pCO_2·ρ_water with the atmospheric pCO_2 fixed at 400 µatm. Colder water dissolves more CO_2, so C_eq now varies with each sample's temperature rather than being held at a single constant.

The routed headwater boundary uses atmospheric equilibrium evaluated at the mean temperature of the supplied observation set; local exchange uses sample-specific equilibrium. During fold-local diagnosis that supplied set excludes the held-out rows. This boundary assumption differs from a measured upstream CO_2 concentration and should not be interpreted as one. Cross-section visualizations are idealized trapezoids, and the vertical velocity profile is a schematic parabola rather than an ADCP measurement. These display products do not add observational constraints to the metrics.

For a source S and fixed local forcing, the explicit solution is C = [q_A C_in + S + k C_eq]/(q_A + k). A negative numerator is incompatible with a nonnegative concentration in this steady balance. The current routing wrapper catches that infeasibility and clips C to zero, which can leave a nonzero balance residual. Three archived held-out MLP predictions are affected. We retain those archived predictions for traceability, but do not treat clipping as a physically conservative solution or their RMSE as a constraint-validated performance result.

2.3 Spatial filtering and diagnosis of the subgrid residual

Reach-scale transport formulations average over heterogeneity within each reach, and the unresolved contributions appear formally as a residual source-sink term. Studying that term requires an operable definition of the filter width Deltax rather than a qualitative notion of subgrid structure.

We perform spatial coarse-graining on the directed NHDPlus HR flowlines. Flowlines are first ordered by their directed topology (upstream to downstream, from endpoint matching and the NHD flow-direction attribute), then grouped into filter cells as individual flowlines, consecutive pairs, or consecutive groups of four (native, ~2×, and ~4× merges). A whole-reach (study_reach) contraction that preserves all cross-group fine edges produces a cyclic cell DAG and is therefore excluded from formal filter claims (no singleton fallback). Each campaign observation is snapped to its nearest cell at each resolved scale. The reported Deltax is the mean length of the unique spatial cells that contain samples at that scale, giving Deltax ≈ 587 m for the native operator. Figure 1 summarizes the filtering construction.

For each date and filter scale, the headwater boundary is atmospheric equilibrium; sampled upstream states inform downstream diagnostic boundaries along directed paths. Excluding a sample's own concentration from its upstream boundary prevents self-conditioning, but does not by itself prevent dependence on another held-out sample. That separate requirement is addressed by fold-local target reconstruction for the main learners. Filter summaries first use available cell-date diagnostics and then aggregate by spatial cell. There are 39, 32 and 27 sampled spatial cells at the three scales, with 43, 36 and 32 sampled cell-date combinations, respectively. These are distinct supports, not statistically independent replicates: cells share network connections and dates. Sparse synchronous coverage means that a cell-mean summary is not a fully observed instantaneous filtered carbon budget.

At the operator level the construction parallels the coarse-graining used in large-eddy simulation and in learned subgrid parameterization studies (Yuval & O’Gorman, 2020); the analogy is limited to spatial filtering, and S_sgs denotes the residual of the filtered river CO_2 balance at the chosen scale. Once resolved transport and gas exchange are recomputed on the filtered balance, the residual implied by the observations is

Eq. (4): S_sgs = k(C - C_eq) - q_A(C_in - C)

S_sgs is a filter-induced closure residual. It can absorb measurement error, errors from the simplified transport representation, and genuinely unresolved processes. It is not a direct measurement of a single unresolved biogeochemical flux. Its magnitude, structure, and learnability are evaluated below as the aggregation scale changes.

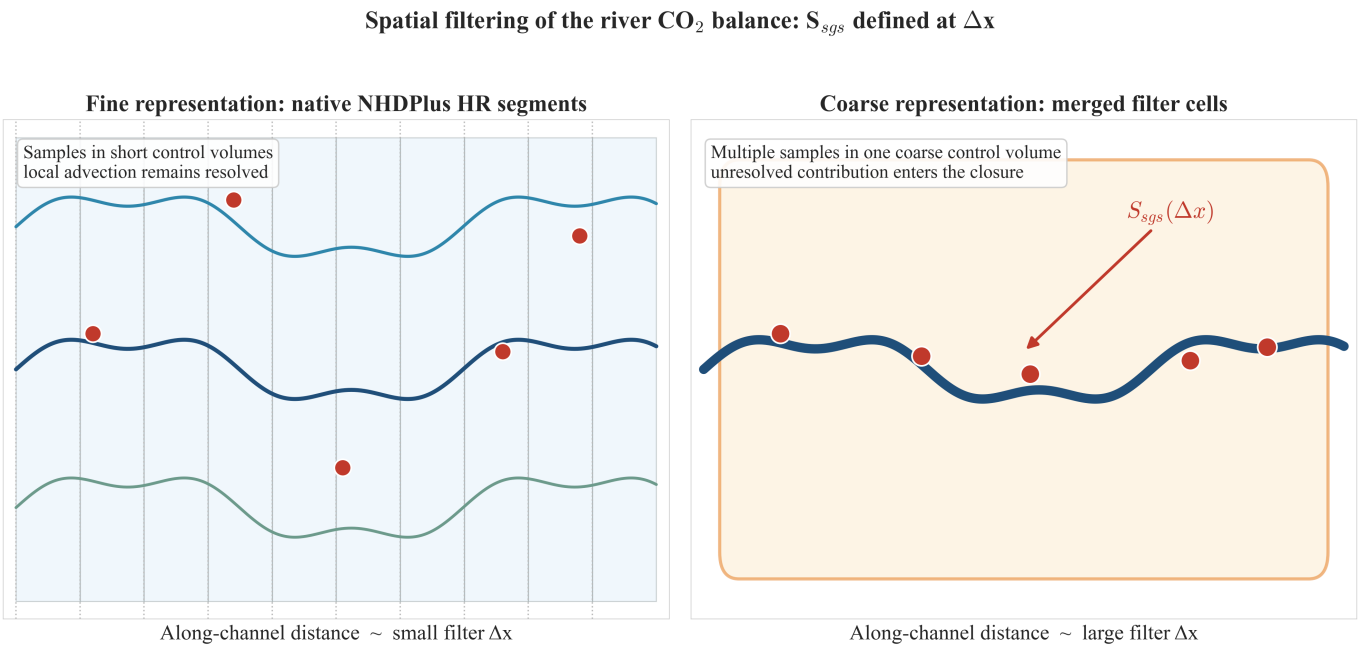


Figure 1. Conceptual representation of spatial coarse-graining of the river CO_2 balance. Left: native NHDPlus HR segments and sample locations within short control volumes. Right: merged filter cells at larger Deltax, for which unresolved contributions to the filtered mass balance are represented by S_sgs.

2.4 Alternative unresolved-process closures

Three closure configurations are compared. They differ only in how S_sgs and k are supplied to Eq. (1).

The Baseline retains the transport and hydraulic formulation of the other configurations while setting S_sgs = 0 and using the energy-dissipation empirical velocity k_emp (Ulseth et al., 2019). It serves as the zero-residual reference for comparison with the alternative closure formulations.

The Residual-AI configuration learns S_sgs from hydraulic and water-quality covariates. Two learners are trained with a fixed seed (42): a multilayer perceptron and a random forest. The candidate predictor pool comprises discharge, velocity, depth, width, slope, temperature, the available carbon chemistry (dissolved organic carbon for 41 of the 120 samples), and the derived dimensionless quantities; fields that are entirely absent from the campaign table are excluded, and missing retained predictors are imputed with medians calculated from the training reaches of each fold. The multilayer perceptron uses hidden layers of 64, 32, and 16 units with learning rate 0.001, early stopping, and fold-specific predictor standardization, and its predicted residual is signed, allowing both source and sink values. The random forest uses 200 trees with maximum depth 12 and no predictor standardization. The training target is the dimensionally consistent residual of Eq. (4), diagnosed from the observed concentrations routed through the directed network (Section 2.3).

The k-correction leaves S_sgs at zero and supplies k_eff = k_emp·exp(g_θ(X)). XGBoost predicts the dimensionless log correction using 300 trees, maximum depth 6, learning rate 0.05 and seed 42. Within each fold, the diagnostic balance gives k_need = q_A(C_in − C_obs)/(C_obs − C_eq), where a finite positive value is required. The training response is log(k_need/k_emp); infeasible inverse targets are excluded rather than replaced by a small positive velocity. In the full-data diagnostic, 72 of 120 targets are infeasible (60%). This fraction describes that diagnostic set, not every training fold or the forward model's domain. The fitted correction still predicts a positive velocity for every held-out sample. None of the 120 archived reach-held-out predictions reaches the log lower bound g = −12. We report the median of the paired ratios k_eff/k_emp, which is not generally the ratio of the two marginal medians.

The diagnosed residual is positive for all 120 samples (a subgrid source; Section 3), yet the Residual-AI learners are left unbounded in prediction. The consequence of this signed, unbounded prediction is examined under held-out evaluation in Section 3.

2.5 Leave-one-reach-out transport-coupled evaluation

The main closure comparison leaves one of the eight logical reaches out at a time. Within each fold, predictor medians and any standardization are fitted on the training reaches; standardization applies to the MLP, but not the tree learners. The fitted closure supplies values for the full network state required by transport. The quasi-steady network is then re-solved, and only predictions for the held-out reach are scored against observed C_aq. The sparse branch is a separate experiment with the limitations in Section 2.8 and should not be assumed to satisfy this complete protocol. No inner hyperparameter-selection loop is used, so the procedure is grouped cross-validation rather than nested cross-validation.

For the main MLP, random-forest and k-correction branches, all concentration labels in the held-out group are removed before training-target diagnosis. Training targets are then joined back by date, reach and sample identifier. Missing predictors are imputed from the training fold, and the fitted closure supplies the full forward network state; only the held-out keys enter scoring. Each key must occur once in the pooled predictions. A held-out-label perturbation check tests whether training targets remain unchanged. Forward headwaters use atmospheric equilibrium, not a fallback to the concentration being scored. Predictor and discharge information remain conditioned on the observed campaign, so this is a closure-transfer experiment rather than an ungauged-basin or future-forcing forecast. Fixed hyperparameters avoid an inner selection step; the MLP's internal early-stopping split is not an additional reach-grouped validation experiment.

R008 contributes 58 samples, whereas three reaches contribute one each. Pooled errors therefore describe the selected samples, not an equally weighted population of reaches. We retain subgroup results (Table 5), distinguish point differences from statistical superiority, and examine the influence of R002. The date-grouped baseline and k-correction runs are time-sensitivity diagnostics, not an inner level of the reach split. The sparse branch retains a different archived target-construction procedure, described explicitly in Section 2.8.

2.6 Metrics and flux diagnostic

The primary metric is the held-out C_aq RMSE in mol m^{-3}. The secondary diagnostic is the area-weighted model flux $\Sigma(F_{CO_2} \cdot A_s)$ in mol d^{-1}, computed from Eq. (3) with the transport-predicted concentration and the transfer velocity of each configuration (k_emp for the Baseline and Residual-AI, k_eff for the k-correction), summed as an overlapping sample-area diagnostic (not a study-area total). An observation-based proxy uses k_emp with observed concentrations. Differences in this area-weighted total across closures indicate how each configuration allocates the balance between sources and gas exchange; the totals are model diagnostics, not independently validated evasion measurements.

2.7 Practical equifinality diagnostic

To characterize compensation between source terms and gas exchange, we define the implied source adjustment

$$S_{implied} = (k_{emp} - k_{eff})(C - C_{eq}) \quad \text{Eq. (5)}$$

At fixed concentration and resolved transport state, S_implied is the source-sink adjustment that makes a model retaining k_emp locally equivalent to a model that uses k_eff and no additional source term. In Eq. (5), C is the observed aqueous concentration at the sample, so the two process allocations are compared at a common observed concentration rather than at their respective forward-model concentrations. A large S_implied together with a small change in concentration error indicates that the observations provide limited discrimination between the two allocations; we refer to this compensating closure behaviour as practical equifinality.

We also calculate a deterministic objective without fitting a learned closure. For a single sample with C_in, q_A and C_eq fixed, $J(k,S) = [C_{obs} - C_{model}(k,S)]^2$ is zero on $S = S_{diag} + (C_{obs} - C_{eq})(k - k_{emp})$. One local concentration equation with two freely varying parameters is therefore

underdetermined. This algebraic result does not establish non-identifiability of an entire routed network or of a restricted closure family. For the pooled diagnostic, all 120 samples share one transfer scale α = k/k_emp and one source S. The objective is an equally sample-weighted SSE, and J_prof(α) = min_S J(α,S) selects the best common source at each α. Diagnosed observational upstream concentrations remain fixed as α and S vary; the network is not rerouted at each grid point.

The profile describes sensitivity under that parameterization. It is neither held-out prediction nor a likelihood, and the code's 10% increase above minimum RMSE is a descriptive display threshold, not a confidence boundary. Model-specific median parameters shown on the surface are reference summaries of heterogeneous predictions, not operating points that reproduce each model's network RMSE. The current loader reconstructs q_A through a residual identity because merged physical columns receive suffixes; this agrees numerically with direct routed Q/A_s for the archived data, but should be replaced by an explicit keyed physical-column join before release.

2.8 Sparse dimensionless closure

A final experiment asks whether the diagnosed residual has a compact dimensionless description. The response is S* = S_sgs/(k_emp C_eq), and candidates are Fr, slope, h/W, log10(Re) and log10(Da). The current implementation defines Da = (k_emp/86400)τ/h, with τ = L/u in seconds, so its time bases are consistent. The older statement that Da lacked this conversion and was discarded is not applicable to the archived results. LASSO with fixed penalty 0.05 provides sparse coefficients, following the descriptive use of dimensionless candidates motivated by Xie et al. (2022).

Two fits must be distinguished. A full-data fit to S* supplies the displayed standardized and original-variable equations; it is descriptive, not a held-out physical law. A separate grouped transport experiment fits the dimensional S_sgs target and reinserts its predictions into the transport balance. The archived sparse branch, unlike the main learners, builds residual targets and predictor medians before splitting the data. Although its scaler and LASSO are refitted by fold, training targets can retain routed dependence on held-out concentrations and preprocessing uses held-out predictor information. Its reported reconstruction R^2 and transport RMSE are therefore provisional diagnostics, not strict leakage-controlled validation. They are retained to document the existing experiment, pending fold-local target reconstruction and imputation. Compactness alone supplies no evidence of predictive utility.

3. Results

3.1 Lower pooled MLP error does not establish stable closure transfer

The archived leave-one-reach-out predictions give a lower pooled point RMSE for the MLP than for the Baseline, but a higher RMSE for the random forest (Tables 2 and 3; Figure 3). Concentration RMSE is 0.0486, 0.0372 and 0.1011 mol m^{-3}, respectively. Corresponding MAE values are 0.0236, 0.0214 and 0.0512 mol m^{-3}, and biases are −0.0236, −0.0055 and +0.0342 mol m^{-3}. Thus the MLP reduces the pooled negative bias, whereas the random forest has a positive bias. These are descriptive out-of-fold metrics, not in-sample fit scores or evidence of a statistically established performance advantage. The baseline date-grouped calculation gives the same 0.0486 mol m^{-3} RMSE because no baseline closure is learned from a split.

Subgroup results distinguish the learners (Table 5; Figures 4 and S1). On R008, RMSE is 0.0278 mol m^{-3} for the Baseline, 0.0248 for the MLP and 0.0239 for the random forest. Across R002–R005, it is 0.0635, 0.0470 and 0.1421 mol m^{-3}, respectively. The MLP therefore has a lower pooled error in both groups; the random forest improves on R008 but deteriorates across the multi-sample tributaries. R002 contains three strongly supersaturated samples (pCO_2 approximately 5455–6066 µatm) and contributes 93.8% of the total squared-error reduction from Baseline to MLP. Removing those three rows from the archived predictions gives RMSE 0.0268 versus 0.0256 mol m^{-3}. This is a scoring sensitivity, not a retraining experiment. Taking the square root of mean reach-specific MSE gives 0.0932 versus 0.0640 mol m^{-3}; equal reach weighting does not remove R002's influence. MLP errors increase on R001, R004 and R007. The scatter therefore documents uneven transfer rather than a uniformly better closure.

The concentration comparison must be read with the process diagnostics. MLP and random-forest flux sums are 1.94×10^{6} and 7.28×10^{5} mol d^{-1}, compared with 1.21×10^{3} for the Baseline (Table 2). Their magnitude alone does not prove physical impossibility because independent evasion measurements and non-overlapping support are absent. A separate numerical failure is directly observable: three held-out MLP states on R005—GC-1 on 3 August, and GC-14 and GC-2 on 6 August—are clipped to zero after an infeasible negative solution. Clipping changes the solved balance. The lower MLP point RMSE is consequently retained as an archived calculation, not a physically validated success. Spatial-support correction and an explicit feasibility treatment must precede a stronger transfer claim.

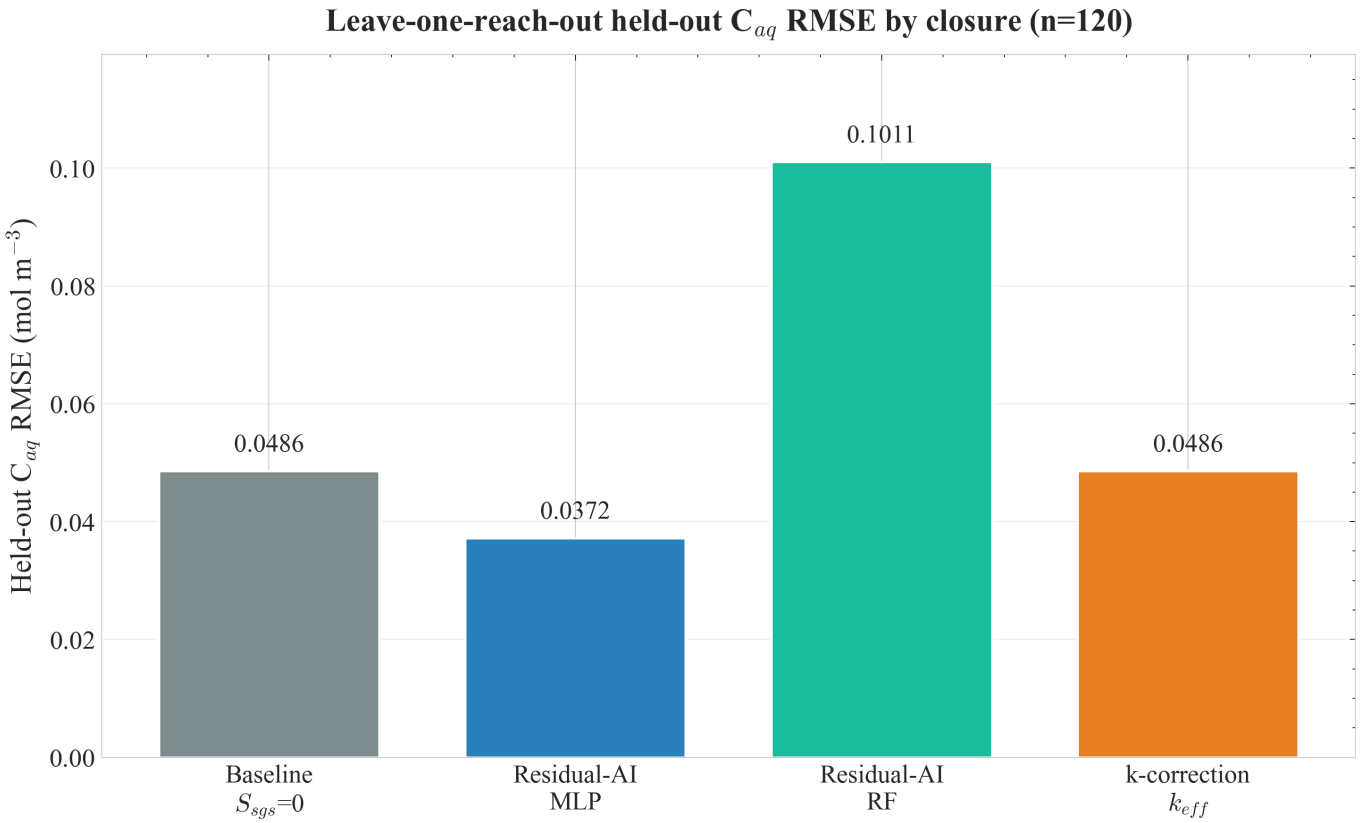


Figure 3. Sample-weighted leave-one-reach-out concentration RMSE for 120 archived predictions. The MLP point RMSE is lower than the Baseline; the random forest and k-correction do not improve it. The MLP result includes three clipped held-out states, and all results remain conditional on the archived spatial support. Bars are point metrics, not uncertainty intervals.

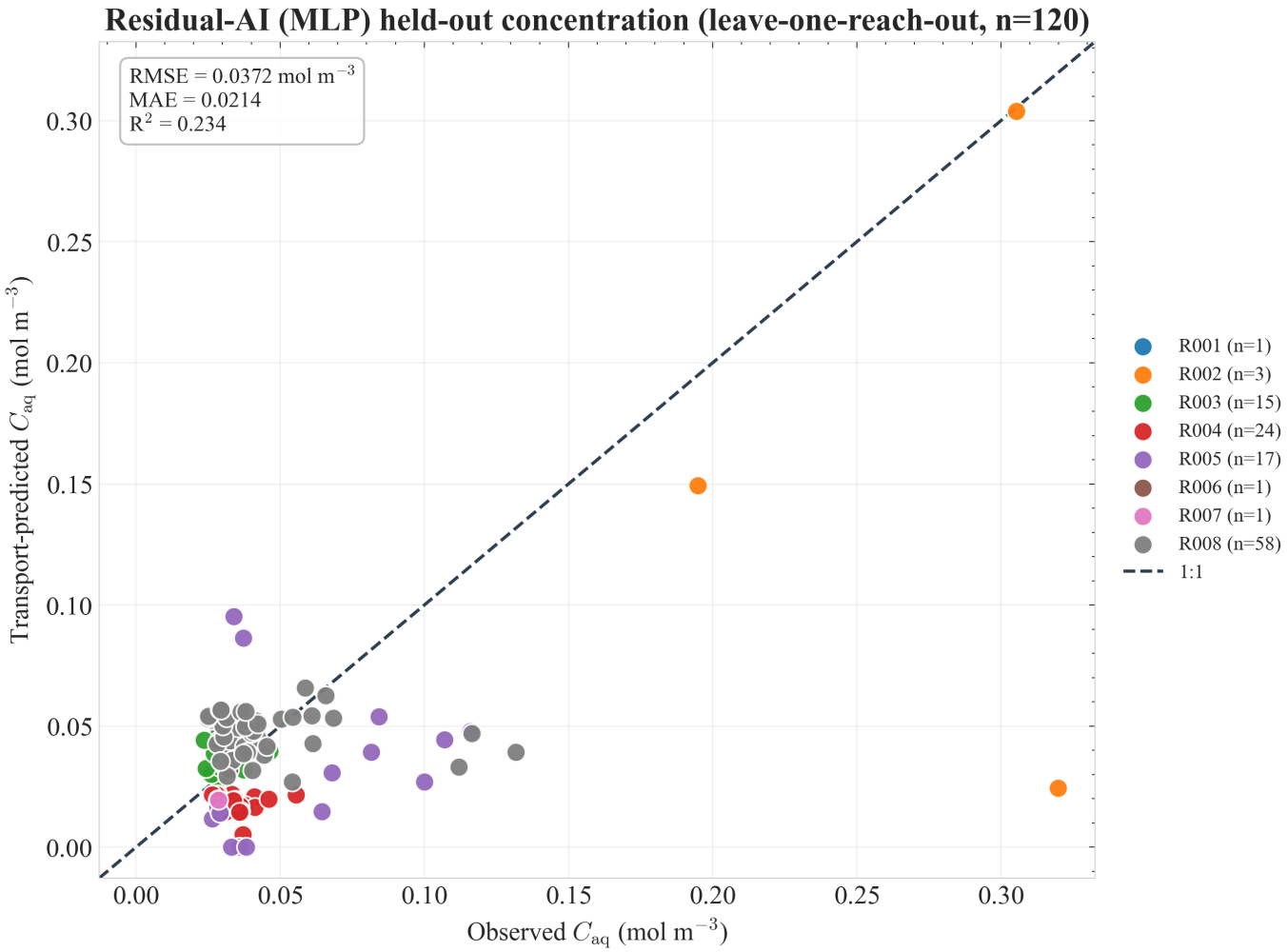


Figure 4. Observed versus transport-predicted held-out C_aq for the Residual-AI (MLP) closure under leave-one-reach-out grouped evaluation (n=120). Colors denote logical reaches; the dashed line indicates 1:1 agreement.

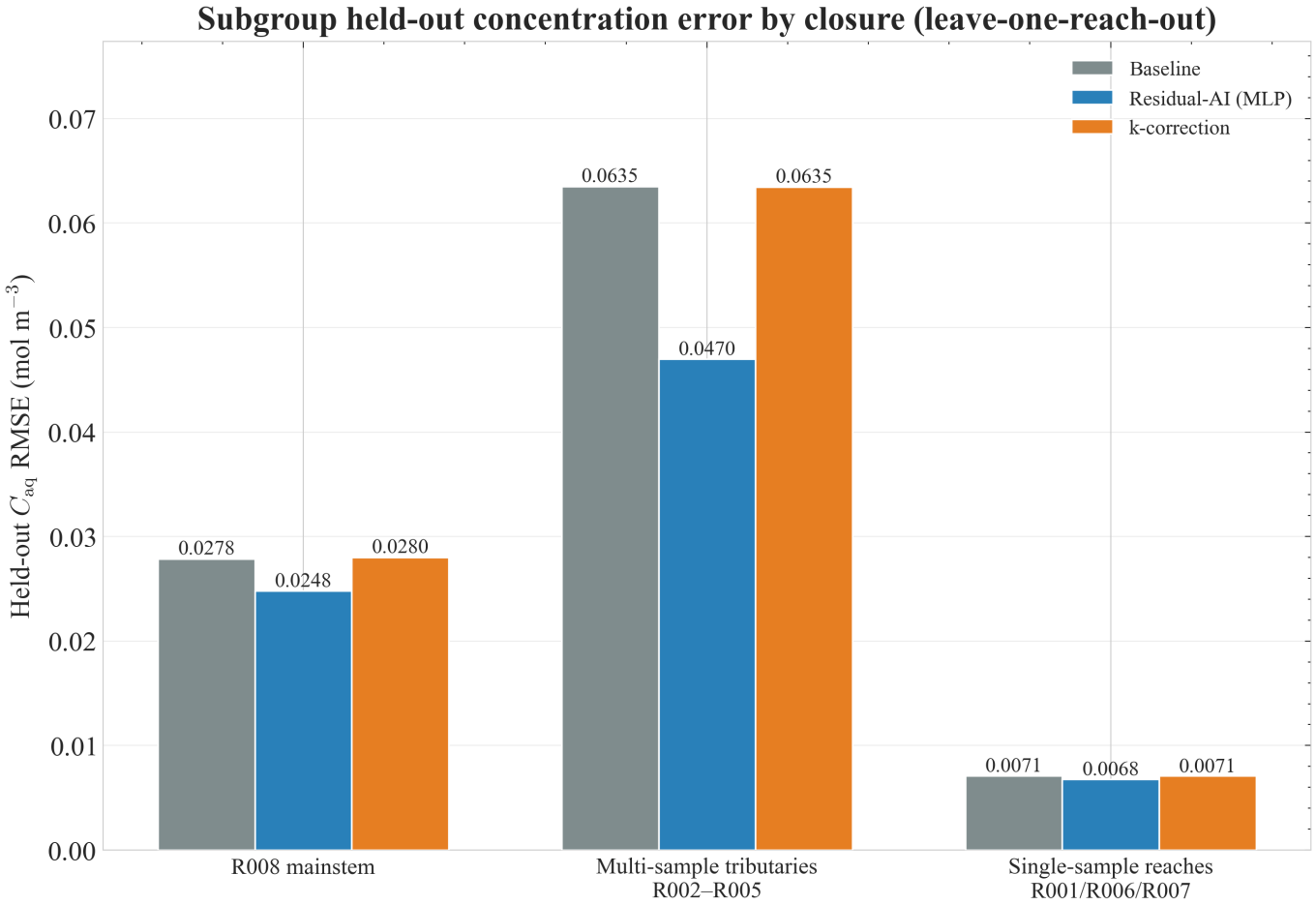


Figure S1. Subgroup C_aq RMSE for the Baseline, Residual-AI (MLP), and k-correction. R008 contains 58 samples; the multi-sample tributary group R002–R005 contains 59 samples; R001, R006, and R007 are single-sample logical reaches.

3.2 A corrected gas-transfer velocity does not improve concentration error

The k-correction gives RMSE 0.048589 mol m⁻³, compared with 0.048574 for the Baseline; both round to 0.0486 (Tables 2 and 3). It therefore does not improve the archived concentration point metric. The full-data inverse diagnostic lacks a positive k-only solution for 72 samples, but the learned forward correction remains defined for all 120 held-out rows. No held-out log correction reaches the imposed lower bound. Median k_eff is 123.3 m d⁻¹, compared with median k_emp of 194.0 m d⁻¹, and the median paired ratio is 0.540 (Table 7). These results indicate reduced transfer velocities without the near-zero saturation asserted in earlier text. Target infeasibility restricts the information available to training; it does not, by itself, prove why the learned correction fails to reduce error.

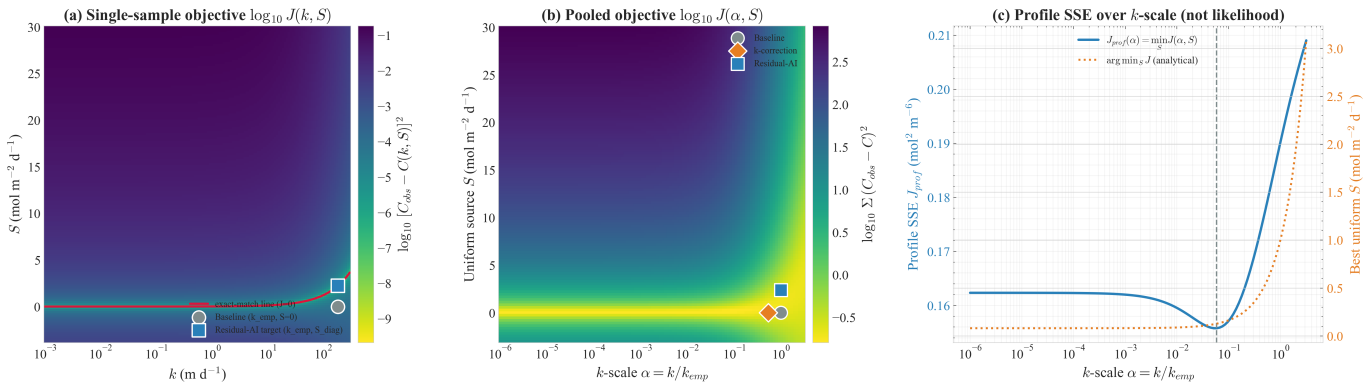


Figure 5. Conditional deterministic source–exchange diagnostic. (a) Single-sample objective with fixed upstream concentration; the line marks the analytic zero valley. (b) Equally sample-weighted pooled SSE for one shared transfer scale and one shared source. Model symbols are median-parameter reference summaries, not equivalent network operating points. (c) Profile SSE after optimizing the shared source at each scale; RMSE spans 0.0360–0.0396 mol m⁻³ over α=0.01–1. The plot also displays smaller scales outside that reporting interval. This is neither a likelihood nor an uncertainty interval, and upstream concentrations are not rerouted over the surface.

That concentration error does not discriminate the k-correction from the Baseline motivates the model-free objective surface examined in Section 3.6 and Figure 5.

3.3 Similar concentration errors coexist with different flux diagnostics

The overlapping sample-area flux diagnostic is 1.21×10³ mol d⁻¹ for the Baseline and 1.26×10³ for the k-correction. Thus the correction slightly increases this diagnostic despite reducing the median transfer velocity: the predicted concentration gradient also changes. The MLP and random forest give 1.94×10⁶ and 7.28×10⁵ mol d⁻¹ (Table 2). These differences document model-dependent allocations under the assigned support, not measured differences in watershed evasion. Without independent evasion observations, their ranking does not determine which allocation is physically correct.

At the common observed concentration, the mean implied adjustment S_implied is 1.013 mol m⁻² d⁻¹ and the mean MLP source prediction is 4.035 mol m⁻² d⁻¹. Their descriptive Spearman correlation is +0.213; no independence-based significance claim is made. Positive source adjustment and reduced transfer velocity can compensate in the local balance, but their differing magnitudes and weak rank association do not make the two fitted closures interchangeable. Table 7 records their transfer and concentration summaries, and Figure S2(c) shows the sample-level relation between the paired transfer ratio and the k-correction flux diagnostic.

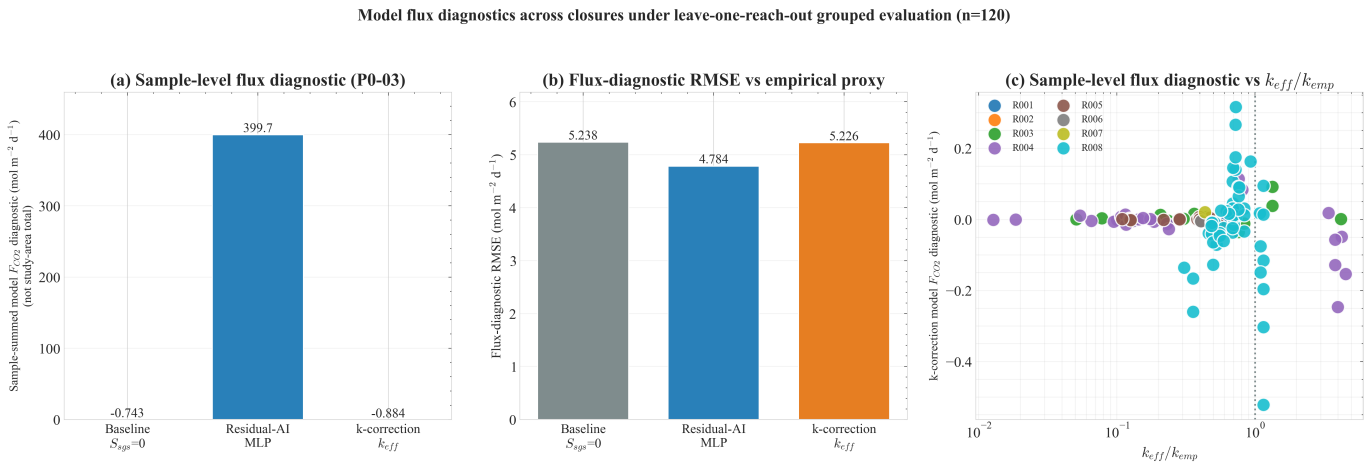


Figure S2. Archived flux-density diagnostics for 120 reach-held-out predictions. (a) Unweighted sum of sample flux densities, approximately −0.743, 399.7 and −0.884 mol m⁻² d⁻¹ for Baseline, MLP and k-correction. This is not the area-weighted diagnostic in Tables 2, 3 and 7; summing densities over samples has no spatial-total interpretation. (b) Flux-density RMSE against an empirical observation-based proxy, not independent flux measurements. (c) Paired transfer ratio versus predicted flux density. Negative density indicates uptake under the model sign convention.

3.4 Filter-scale residual on resolved coarsenings

On the resolved path-connected coarsenings (native, ~2×, ~4×), cell-mean |S_sgs| remains near 3.31–3.57 mol m⁻² d⁻¹ (Deltax ≈ 587–1010 m; Table 6; Figure 6). Cell-level variance stays of the same order (about 17–20). A study-reach whole-reach merge is excluded: preserving all cross-group fine edges yields a cyclic contracted DAG, so the reach assignment is not a verified DAG compression of the fine NHD graph and cannot support a formal scale-invariance claim through that aggregation. Within the resolved scales the near-stability of |S| under sparse synchronous coverage is a sampling-conditional sensitivity of the conserved operator, not a universal scaling law.

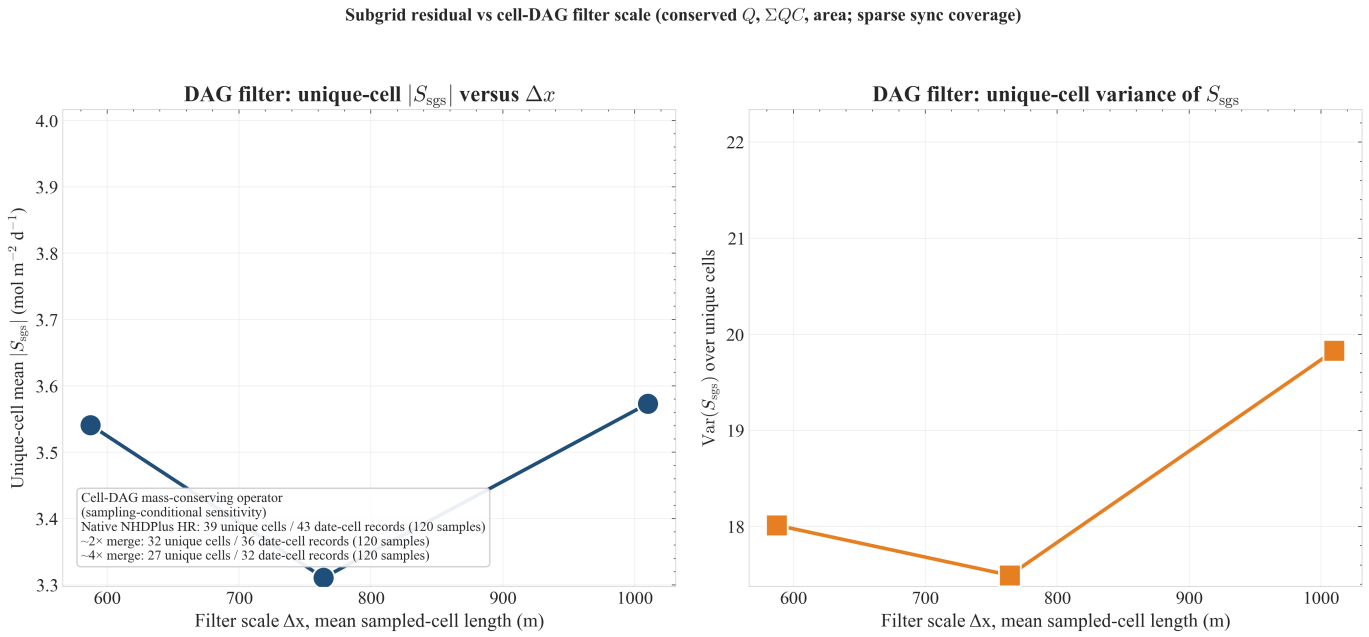


Figure 6. Filter-scale dependence of the diagnosed residual on resolved scales only (native, ~2×, ~4×; study_reach excluded for contraction cycle). Left: cell-mean |S_sgs|. Right: cell-level variance of S_sgs. Cell-mean |S| stays near 3.3–3.6 mol m⁻² d⁻¹ under sparse synchronous coverage; this is a sampling-conditional sensitivity, not a claim of scale-invariance through the study-reach aggregation.

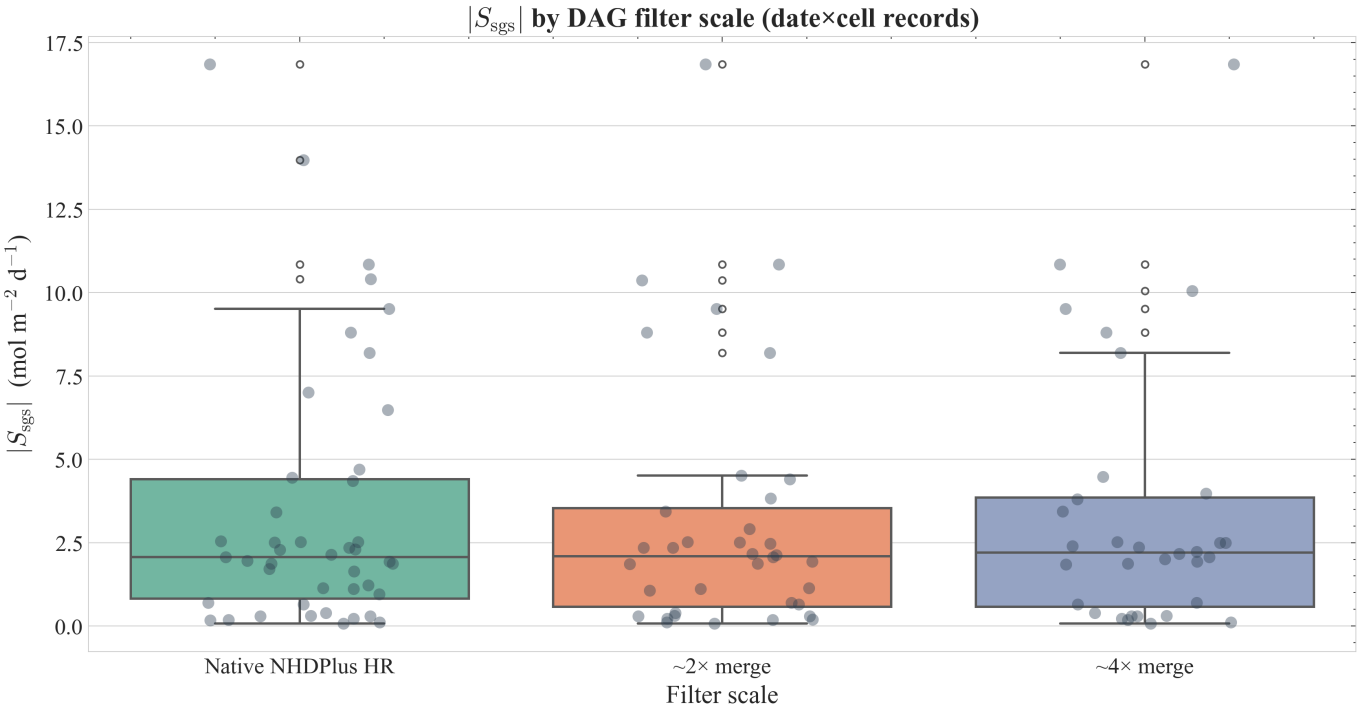


Figure S3. Distribution of |S_sgs| across sampled cell-date records at resolved scales, with 43, 36 and 32 records. This differs from the unique-spatial-cell summary in Figure 6 and Table 6. Connected cells and repeated dates are not statistically independent replicates. Study-reach aggregation is excluded because preserving all cross-group edges produces a cyclic contraction.

3.5 A compact sparse description has not passed strict held-out validation

The full-data LASSO refit retains three of five candidate terms (Table 8; Figure 7). With subscript z denoting standardization by full-data mean and standard deviation, its descriptive equation is

$$S^* \approx 1.063 - 0.08591 \cdot \mathrm{Fr}_z - 0.3123 \cdot \mathrm{Slope}_z + 0.3457 \cdot \log10(\mathrm{Da})_z$$

Froude number and slope have negative coefficients, while log10(Da) has a positive coefficient; relative depth and log10(Re) shrink to zero. The archived grouped S* reconstruction gives $R^2 = -0.448$. The distinct dimensional-target transport LASSO gives concentration RMSE 0.0529 mol m^{-3}, compared with Baseline 0.0486 (Table 9). Neither number establishes a validated sparse closure: the target and imputation leakage described in Section 2.8 must first be removed. The full-data equation is a compact summary of the current diagnosis, not an independently identified process law or the single coefficient vector used in every held-out transport fold.

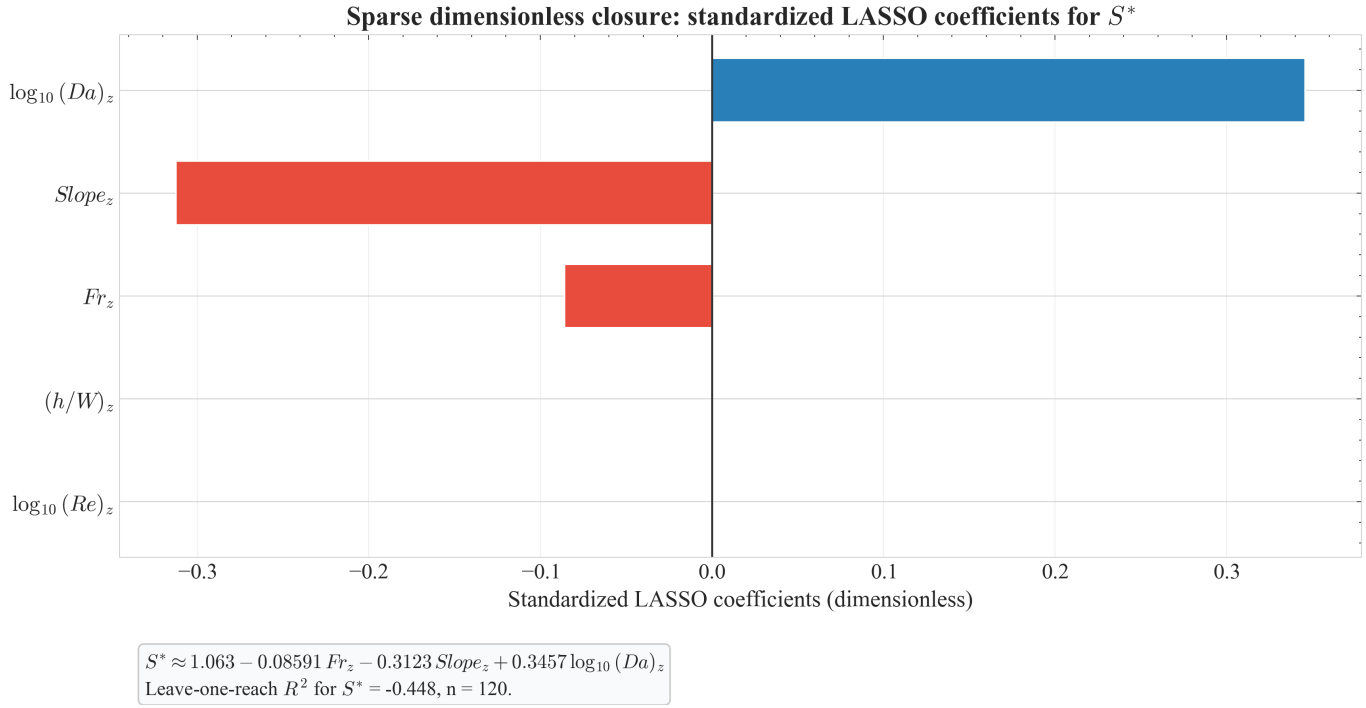


Figure 7. Full-data standardized LASSO coefficients for S*. The retained terms are Fr_z, Slope_z and log10(Da)_z. The displayed grouped reconstruction R² of −0.448 is provisional because targets and imputation are not fully fold-local in this branch; the plot is retained as an archived descriptive result, not a validated law.

3.6 Conditional objective surface and profile SSE

The single-sample surface reproduces the analytic zero valley of the fixed-boundary equation (Figure 5). Its flat profile follows from allowing a separate source value at every transfer velocity; it is not an empirical discovery about all possible concentration measurements. Pooling the 120 samples with a common α and source gives a non-flat profile. RMSE ranges from 0.0360 to 0.0396 mol m^{-3} over α = 0.01–1, with the archived minimum near α = 0.05843 and S = 0.1277 mol m^{-2} d^{-1}. This describes limited change in the selected deterministic error metric across that interval. Its interpretation depends on fixed observational upstream boundaries, sample weighting, shared parameters and the current spatial support. It cannot be compared directly with held-out network RMSE as if it were another validated predictive model.

4. Discussion

4.1 Concentration skill and physical validity answer different questions

With fold-local targets, the MLP lowers the archived sample-weighted concentration RMSE; the random forest does not. The MLP gain is concentrated in R002 rather than shared uniformly across reaches, and three held-out states require clipping. These observations separate numerical concentration fit from a physically admissible and stable closure. This distinction follows the purpose-dependent evaluation advocated by Bennett et al. (2013) and the treatment of discrepancy in Vilas et al. (2023). The subgroup pattern is also model-specific: the MLP improves pooled errors on both R008 and R002–R005, whereas the random forest deteriorates on the tributary group. A single statement that residual learning either succeeds or fails would conceal that difference.

Two checks are needed before interpreting the MLP comparison more strongly. First, support lengths and routing must correspond to a declared physical domain. In the archived length diagnostic, R002's assigned control lengths total 837.76 m against a recorded NHD support of 170.22 m. That discrepancy is especially consequential because R002 dominates the error reduction; neither recorded number should be accepted as the true domain without path verification. Second, source predictions must yield admissible states without hiding conservation errors. A feasibility-aware bound or constrained solve should be defined from the current routed state, not selected to improve the held-out score. Reporting

source predictions, raw concentrations, applied corrections and balance residuals would make this check auditable. These repairs change the computation, so their eventual metrics cannot be anticipated by editing the current prose.

The Baseline is a reference configuration, not a demonstrated physical truth. A learned closure can improve concentration error while assigning an unsupported process contribution; conversely, a poor closure does not prove that unresolved sources are absent. The conditional surface in Figure 5 explains local compensation, but does not show that concentration data can never distinguish processes.

4.2 Process allocation and practical equifinality

For 60% of full-data diagnostic samples, no positive k_need matches the observed concentration with zero residual source under the assigned boundary and transport state. This is an inverse-target feasibility statement. It does not make the learned forward model undefined: positive k_eff predictions exist for every held-out sample, and none reaches the lower bound. Their failure to improve RMSE should therefore be reported separately from target infeasibility. The balance nevertheless explains why an added source and reduced gas exchange can offset one another at fixed concentration. That compensation motivates a diagnostic of process allocation, rather than a causal explanation inferred from a small change in aggregate error.

Here, practical equifinality is used narrowly for source–exchange compensation under the stated conditional calculation. The single-sample zero valley is an analytic benchmark; the pooled profile adds a data-dependent sensitivity under one shared parameterization. Neither is a formal structural-identifiability analysis or an error-model-based likelihood assessment. The large differences in learned-model flux diagnostics are also not, by themselves, evidence of equifinality. They show that changing the closure changes the inferred process allocation. Reach weighting, the assumed common source, and spatial-support repair may change the pooled profile; until those sensitivities are checked, the reported RMSE interval should remain descriptive. Independent gas-transfer or source constraints would address a different question from reducing concentration error alone.

This process-allocation ambiguity concerns the closure form; the filtering experiment addresses a related but distinct source of variability in the diagnosed residual.

4.3 What filtering and sparse representation reveal about the residual

On resolved filter scales only, cell-mean |S_sgs| stays near 3.3–3.6 mol m^{-2} d^{-1} from native NHDPlus HR resolution through ~4× merges. Study-reach aggregation is excluded after an unresolved contraction cycle, so the manuscript does not claim scale-invariance through that coarser operator. Within the resolved range the result is a sampling-conditional sensitivity under sparse synchronous coverage, bounded by the implemented flow-directed operator (Yuval & O’Gorman, 2020, analogy limited to spatial filtering).

The sparse description retains Froude number, slope and log10(Da), not a two-term relation. Its archived reconstruction R^2 is negative, and its dimensional transport error exceeds the Baseline. However, incomplete fold isolation prevents using these results as a definitive test of sparse closure transfer. Repairing the target construction and imputation is necessary even for a negative result: leakage does not become acceptable because a model performs poorly. The full-data coefficients can still document associations in the diagnosed residual, provided that coefficient scaling, response definition and validation status are stated explicitly.

4.4 Implications for environmental-model evaluation

This case separates three questions: whether concentrations are predicted more accurately, whether the numerical states satisfy the specified balance, and whether the allocated processes have independent observational support. A lower RMSE answers only the first. The comparison should therefore retain concentration errors, feasibility and conservation checks, and process diagnostics as separate outputs rather than combine them into a single claim of physical improvement.

The present conclusions are bounded by strongly unequal reach support, the hydraulic-geometry representation of channel width and depth, incomplete covariates (alkalinity, nitrogen, phosphorus, photosynthetically active radiation), and the absence of independent evasion measurements. The Water Quality Portal merge and the StreamPULSE search returned no usable additional constraints for this campaign. These limitations restrict inference to the East River experiment, and they also identify the observations that would most help to discriminate closures: more balanced reach sampling, better-resolved channel geometry, and independent constraints on gas exchange. The central implication is that lower concentration error alone is insufficient to determine which allocation of unresolved processes is better supported.

5. Conclusions

The archived reach-held-out calculation gives MLP concentration RMSE 0.0372 mol m^{-3}, compared with 0.0486 for the Baseline and 0.1011 for the random forest. The MLP point reduction is concentrated in R002 and includes three clipped states; it is not a validated generalization advantage. The k-correction also rounds to 0.0486 and does not improve the point metric. Separately, 60% of the full-data inverse targets lack a positive k-only solution. A fixed-boundary single-sample balance admits exact source–exchange compensation, while the pooled deterministic profile describes a conditional, parameterization-dependent sensitivity. These findings do not establish general network non-identifiability. Filter-scale interpretation is restricted to the resolved coarsenings, and the sparse branch remains provisional until its targets and preprocessing are made fold-local.

The analysis combines spatial filtering, transport-coupled grouped prediction and a conditional objective-surface diagnostic without treating them as interchangeable evidence. Public measurements provide the observations; the present code supplies the model predictions and residual calculations. The immediate contribution is an auditable East River comparison and a clear boundary between concentration skill and process attribution. Correcting spatial support, numerical feasibility and sparse-fold isolation is necessary before a final empirical interpretation. No watershed evasion total, independent flux validation, universal filter law or transfer to another basin is established by the current results.

6. Data availability

The water-chemistry and pCO_2 observations are public data from Saccardi and Winnick (2021), archived in HydroShare resource 9f907b46baa848e180c49339d605bf31. The DIC supplement, network shapefiles and hydraulic tables are associated with resource 2a2132999fb84214aad0596783812db2. Outlet discharge is from USGS gage 09112500, and hydrographic geometry uses NHDPlus HR for HUC 14020001. These inputs are third-party observations and published relations; they are not claimed as new measurements by the present authors. Model predictions, diagnosed residuals, evaluation metrics and plots are derived calculations. The code location provided for this project is https://github.com/Coucou2016/river-carbon-transport; this local review does not certify that the remote repository contains the current working-tree version. A version-specific release, dependency record, raw-input hashes, selection log and complete artifact manifest are required at submission. The local review package records the inspected snapshot and the scope of successful recomputation, rather than claiming a complete fresh raw-to-publication pipeline run.

Tables

Table 1. East River study network: logical reaches and mainstem confluences (logical reporting / y-threshold simplified tree, not a verified NHD directed compression).

Reach	Stream name	Role	Joins mainstem	Legacy length (km)	Mapped NHD lines	Samples	Note
R001	Bradley Creek	Tributary	R008_S4	12.6	0	1	Tributary headwater; n = 1, schematic only
R002	Bradley Meadow	Tributary	R008_S4	12.5	15	3	Physical support unresolved
R003	Rock Creek	Tributary	R008_S3	6.1	29	15	Multi-sample tributary
R004	Copper Creek	Tributary	R008_S2	9.9	140	24	Mapping is not path validation
R005	Gothic Creek	Tributary	R008_S3	4.8	6	17	Multi-sample tributary
R006	Quigley Creek	Tributary	R008_S6	2.8	8	1	Single sample; schematic
R007	Rustlers Gulch	Tributary	R008_S5	23.5	2	1	Single sample; schematic
R008	East River mainstem	Mainstem (S1–S6)	outlet	20.2	193	58	Six logical routing segments

Mapped counts come from gis_reach_line_mapping.csv (393 assigned lines); they are not logical-node counts or proof of valid transport paths. Legacy lengths are retained from the original reporting table to preserve its information, but are not certified control-volume domains and must not be used to reconstruct area. For example, the R002 legacy length, the 170.22 m NHD support record and the 837.76 m control-length sum describe inconsistent supports. A verified path-domain table must replace this legacy-length field before submission.

Table 2. Archived reach-held-out point metrics. C RMSE is in mol m^{-3}; F diagnostic is the overlapping sample-area sum in mol d^{-1}; the transfer ratio is the median of paired ratios. Sparse-LASSO validation is provisional, and the MLP includes three clipped states. “Yes” refers only to lower archived point RMSE.

Scheme / model	C RMSE	F diagnostic	k_eff/k_emp	Lower point RMSE?
Baseline	0.0486	1.21×10^{3}	1.00	—
k-correction / XGBoost	0.0486	1.26×10^{3}	0.540	No
Residual-AI / MLP	0.0372	1.94×10^{6}	1.00	Yes
Residual-AI / random forest	0.1011	7.28×10^{5}	1.00	No
Sparse-Π / LASSO	0.0529	1.57×10^{6}	—	No

Table 3. Primary concentration metrics and secondary flux diagnostics. C errors have units mol m^{-3}; F RMSE and bias have units mol m^{-2} d^{-1} against an empirical proxy; F total denotes the overlapping sample-area diagnostic in mol d^{-1}, not watershed flux. R^2 is dimensionless. Values describe archived predictions, subject to the limitations in Section 3.1.

Scheme / model	C RMSE	C MAE	C Bias	C R2	F RMSE	F Bias	F total	n
Baseline	0.0486	0.0236	-0.0236	-0.306	5.238	-3.205	1.21×10^{3}	120
k-correction / XGBoost	0.0486	0.0236	-0.0236	-0.307	5.226	-3.207	1.26×10^{3}	120
Residual-AI / MLP	0.0372	0.0214	-0.0055	0.234	4.784	0.131	1.94×10^{6}	120
Residual-AI / random forest	0.1011	0.0512	0.0342	-4.654	5.935	0.017	7.28×10^{5}	120

Table 4. Evaluation-status summary; in-sample concentration metrics are not used as predictive evidence.

Branch	Archived evidence	Remaining qualification
Main MLP / random forest	Fold-local residual targets; 120 OOF keys each	Spatial support unresolved; MLP has 3 clipped OOF states
k-correction	Fold-local feasible inverse targets; 120 OOF keys	72 full-data targets infeasible; no OOF floor saturation
Sparse LASSO	Full-data coefficient description and grouped diagnostic	Target reconstruction and imputation not fully fold-local
Profile SSE	Fixed-boundary deterministic sensitivity	No network rerouting, likelihood or confidence interval

Table 5. Subgroup point metrics under reach-held-out evaluation. Residual-AI denotes the MLP in this table. C RMSE has units mol m^{-3}; evidence-weight labels identify reporting strata, not formal statistical weights.

Scheme	Subgroup	Evidence weight	C RMSE	C R2	n
Baseline	All 120 samples	all	0.0486	-0.306	120
Baseline	R008 East River mainstem	mainstem	0.0278	-0.751	58
Baseline	Multi-sample tribs R002-R005	tributary	0.0635	-0.271	59
Baseline	One-sample reaches (schematic) R001/R006/R007	schematic	0.0071	—	3
k-correction	All 120 samples	all	0.0486	-0.307	120
k-correction	R008 East River mainstem	mainstem	0.0280	-0.770	58
k-correction	Multi-sample tribs R002-R005	tributary	0.0635	-0.269	59
k-correction	One-sample reaches (schematic) R001/R006/R007	schematic	0.0071	—	3
Residual-AI	All 120 samples	all	0.0372	0.234	120
Residual-AI	R008 East River mainstem	mainstem	0.0248	-0.389	58
Residual-AI	Multi-sample tribs R002-R005	tributary	0.0470	0.305	59
Residual-AI	One-sample reaches (schematic) R001/R006/R007	schematic	0.0068	—	3

Table 6. Filter-scale diagnostics after mapping 120 observations to coarsened NHDPlus HR cells. Sampled cells are unique spatial supports, not independent replicates. Mean |S_sgs| has units mol m^{-2} d^{-1} and variance has squared units. The corresponding cell-date counts are 43, 36 and 32.

Scale	Δx (m)	Cells	Sampled cells	Samples	Cell-mean |S_sgs|	Cell Var(S_sgs)
Native NHDPlus HR	587	536	39	120	3.541	18.015
~2× merge	764	502	32	120	3.311	17.494
~4× merge	1010	486	27	120	3.573	19.831

Table 7. Transfer and concentration summaries from the reach-grouped protocol. C RMSE is in mol m^{-3}, median k in m d^{-1}, and F total is the overlapping sample-area diagnostic in mol d^{-1}. The ratio column uses paired ratios, not marginal-median division. Residual-AI denotes MLP.

Scheme	C RMSE	F total	Median k	k_eff/k_emp
Baseline	0.0486	1.21×10^{3}	194.0	1.00
Residual-AI	0.0372	1.94×10^{6}	194.0	1.00
k-correction	0.0486	1.26×10^{3}	123.3	0.540

Table 8. Sparse dimensionless closure (Pi-group LASSO).

Item	Result
Standardized-predictor form	$S^* \approx 1.063 - 0.08591\cdot Fr_z - 0.3123\cdot Slope_z + 0.3457\cdot log10(Da)_z$
Original-variable form	$S^* \approx 2.559 - 2.329\cdot Fr - 7.262\cdot Slope + 0.7527\cdot log10(Da)$
Dominant terms	$0.346\cdot log10(Da)_z - 0.312\cdot Slope_z - 0.0859\cdot Fr_z$
Archived grouped R2 for S*	-0.448; provisional, not leakage-controlled validation

Table 9. Archived dimensional-target sparse transport diagnostic, pending fold-local target and imputation repair. This is not strict validation of the full-data S* equation. C RMSE is in mol m^{-3}; F RMSE is in mol m^{-2} d^{-1} against an empirical proxy. Baseline C RMSE is 0.0486.

Scheme	C RMSE	C R2	F RMSE	n
Sparse-Π / LASSO (leave-one-reach)	0.0529	-0.546	4.641	120

References

1. Battin, T. J., et al. (2023). River ecosystem metabolism and carbon biogeochemistry in a changing world. Nature, 614, 676–687. https://doi.org/10.1038/s41586-022-05500-8

2. Bennett, N. D., et al. (2013). Characterising performance of environmental models. Environmental Modelling & Software, 40, 1–20. https://doi.org/10.1016/j.envsoft.2012.09.011

3. Gao, Y., et al. AI cross-fusion approach for river carbon transport. Unpublished manuscript in preparation; bibliographic details and availability require author confirmation. Not used to establish published priority.

4. Gómez-Gener, L., Rocher-Ros, G., et al. (2021). Global carbon dioxide efflux from rivers enhanced by high nocturnal emissions. Nature Geoscience, 14, 647–653. https://doi.org/10.1038/s41561-021-00722-3

5. Hotchkiss, E. R., et al. (2015). Sources of and processes controlling CO_2 emissions change with the size of streams and rivers. Nature Geoscience, 8, 696–699. https://doi.org/10.1038/ngeo2507

6. Markovich, K. H., White, J. T., & Knowling, M. J. (2022). Sequential and batch data assimilation approaches to cope with groundwater model error: An empirical evaluation. Environmental Modelling & Software, 156, 105498. https://doi.org/10.1016/j.envsoft.2022.105498

7. Raymond, P. A., et al. (2012). Scaling the gas transfer velocity and hydraulic geometry in streams and small rivers. Limnology and Oceanography: Fluids and Environments, 2, 41–53. https://doi.org/10.1215/21573689-1597669

8. Saccardi, B., & Winnick, M. J. (2021). Improving predictions of stream CO_2 concentrations and fluxes using a stream network model: A case study in the East River Watershed, CO, USA. Global Biogeochemical Cycles, 35, e2021GB006972. https://doi.org/10.1029/2021GB006972

9. Vilas, M. P., et al. (2023). TALKS: A systematic framework for resolving model-data discrepancies. Environmental Modelling & Software, 163, 105668. https://doi.org/10.1016/j.envsoft.2023.105668

10. Xie, X., Samaei, A., Guo, J., Liu, W. K., & Gan, Z. (2022). Data-driven discovery of dimensionless numbers and governing laws from scarce measurements. Nature Communications, 13, 7562. https://doi.org/10.1038/s41467-022-35084-w

11. Yuval, J., & O'Gorman, P. A. (2020). Stable machine-learning parameterization of subgrid processes for climate modeling at a range of resolutions. Nature Communications, 11, 3295. https://doi.org/10.1038/s41467-020-17142-3

12. Ulseth, A. J., Hall, R. O., Jr., Boix Canadell, M., Madinger, H. L., Niayifar, A., & Battin, T. J. (2019). Distinct air–water gas exchange regimes in low- and high-energy streams. Nature Geoscience, 12, 259–263. https://doi.org/10.1038/s41561-019-0324-8